

Module 02: Random Access Protocols: Pure & Slotted ALOHA

AKTU BCS603 Focus: Contention resolution, Vulnerable time derivations, throughput equations, Poisson traffic model, and solved numerical problems.

Random Access MAC Protocols: Pure ALOHA vs Slotted ALOHA

1. Pure ALOHA (Unsynchronized / Continuous)

Vulnerable Time = $2 \cdot T_{fr}$

Vulnerable Window: $[t - T_{fr}, t + T_{fr}]$

$S = G \cdot e^{-2G}$

Max Throughput $S_{max} = 1 / (2e) \approx 0.184$ (18.4%) at $G = 0.5$

- Station transmits immediately whenever data is ready.
- Highest collision probability; completely uncoordinated.

Even a 1-bit overlap destroys both transmitting frames!

2. Slotted ALOHA (Discrete Clock Synchronized)

Vulnerable Time = $1 \cdot T_{fr}$

Vulnerable Window: Exactly 1 Slot (T_{fr})

$S = G \cdot e^{-G}$

Max Throughput $S_{max} = 1 / e \approx 0.368$ (36.8%) at $G = 1.0$

- Time divided into discrete slots equal to frame duration T_{fr} .
- Station **MUST** wait for beginning of next slot to transmit.

Doubles throughput capacity compared to Pure ALOHA!

ALOHA Mathematical Evaluation & Poisson Distribution Metrics

Traffic Load (G): Average number of frames generated by the whole system per frame transmission time (T_{fr}).

Probability of k Frames: $P(k) = (G^k \cdot e^{-G}) / k!$ Probability of zero frame collisions $P(0) = e^{-G}$ for 1 slot; e^{-2G} for 2 slots.

AKTU Numerical Formula: Idle Slots Probability = e^{-G} . If 10% slots idle: $e^{-G} = 0.10 \rightarrow G = \ln(10) \approx 2.3$. Throughput $S = G \cdot e^{-G} = 2.3 \times 0.1 = 0.23$.

1. ALOHA Mathematical Summary

Protocol	Vulnerable Time ($V_{t\$}$)	Throughput Function	Peak Load ($G^{*\$}$)	Maximum Efficiency
Pure ALOHA	$2 \cdot T_{fr}$	$S = G \cdot e^{-2G}$	$G = 0.5$	18.4% ($1 / 2e$)
Slotted ALOHA	$1 \cdot T_{fr}$	$S = G \cdot e^{-G}$	$G = 1.0$	36.8% ($1 / e$)

2. Why Slotted ALOHA Outperforms Pure ALOHA

Pure ALOHA me agar koi dusri station kisi frame ke beech me bhi transmit kar de, to collision ho jata hai. Slotted ALOHA stations ko time boundaries par constrain karta hai: transmissions sirf slot start par ho sakti hain. Isse collision window aadhi ho jati hai ($2T_{fr}$ o T_{fr}), aur channel throughput seedhe double ho jata hai!

AKTU Exam Shortcut: Agar question me Poochha jaye ki Pure ALOHA me $G=0.5$ par throughput kya hai: $S = 0.5 \cdot e^{-1} = 0.5 / 2.718 = 0.184$ (18.4%). Calculator par e^{-x} function use karna na bhoolein!